

# Investigate Hotel Business

## using Data Visualization

Exploratory Data Analysis — Final Project Report  
*Understanding Booking & Cancellation Behaviour*

|                    |                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------|
| Dataset            | Hotel bookings, 2017–2019 — 119,390 rows × 29 columns (cleaned to 117,853 rows)                         |
| Tools              | Python — Pandas, NumPy, Matplotlib, Seaborn (Jupyter Notebook)                                          |
| Scope              | Univariate, bivariate, and multivariate analysis across booking, pricing, and cancellation behaviour    |
| Business questions | Hotel-type popularity • Effect of stay duration on cancellations • Effect of lead time on cancellations |

*Maker – SAHIL VERMA*

*Phone Number – 8448752330*

*Mail-ID – [sahilv23910@gmail.com](mailto:sahilv23910@gmail.com)*

*Batch - DA & BA*

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# 1. Project Overview

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This project analyses three years of hotel booking data (2017–2019) covering 119,390 reservations across two properties — a City Hotel and a Resort Hotel — to understand what drives guests to book and what drives them to cancel. The raw dataset was cleaned and standardized to 117,853 usable bookings, then explored through univariate, bivariate, and multivariate visualizations built in Python (Pandas, NumPy, Matplotlib, Seaborn).

Understanding booking and cancellation behaviour matters directly to the hotel's bottom line. A cancelled room is lost or heavily discounted revenue, and cancellations that spike under specific conditions — long lead times, long stays, certain deposit types, or certain booking channels — are largely predictable and therefore preventable. Insight from this data can inform:

- Revenue management — dynamic pricing by season, hotel type, and booking channel.
- Cancellation and deposit policy — where to tighten terms without discouraging genuine bookings.
- Staffing and operations — anticipating demand peaks and troughs by month.
- Guest retention — identifying which customer segments are most reliable and worth investing in.

## 1.1 Business Objective

The single deliverable of this project is a data-driven analysis that answers three core business questions, translated into concrete, actionable recommendations for hotel management:

- **Hotel-type popularity** — which hotel type (City or Resort) is booked most often, and how does demand vary through the year?
- **Stay duration** — does the length of stay affect the probability that a booking is cancelled?
- **Lead time** — does the gap between booking date and arrival date affect the probability of cancellation?

The goal is not just to describe these patterns, but to turn each one into a policy or pricing action a non-technical hotel manager could act on.

## 1.2 Data Preparation Summary

The raw file contained 119,390 rows and 29 columns spanning three years of arrivals. Before analysis, the dataset was assessed and cleaned as follows:

- **Missing values** — children (4 missing) filled with 0; agent (16,340 missing) filled with 0, interpreted as a direct, non-agent booking; city (488 missing) filled as "Unknown"; the company column was dropped entirely as it had no meaningful relationship to the remaining fields.
- **Duplicates** — duplicate rows were kept rather than dropped, since identical hotel, date, room type, and price combinations plausibly represent different guests rather than logging errors.
- **Inconsistent categories** — the "Undefined" meal value was recategorized as Self-Catering for clarity.
- **ADR anomalies** — a single booking with an average daily rate of 5,400 (roughly 10× the next-highest value, and already cancelled) was treated as a data-entry error and dropped; negative ADR values (impossible in practice) were also dropped. Bookings with an ADR of exactly 0 were kept, since they map to explainable causes such as complimentary stays, promotions, or loyalty rewards, and showed a far lower cancellation rate (10.9%) than the dataset average (37%).
- **Zero-guest bookings** — rows where adults, children, and babies were all simultaneously zero were removed, since these cannot represent a real stay.
- **Date handling** — the separate day, month, and year columns were combined into a single arrival\_date field and the originals dropped.

*After cleaning, the working dataset used for all charts below contains 117,853 bookings.*

## 2. Univariate Analysis

Single-variable views that establish the baseline shape of the data before looking at relationships between variables.

### 2.1 — Which hotel type is booked more often?

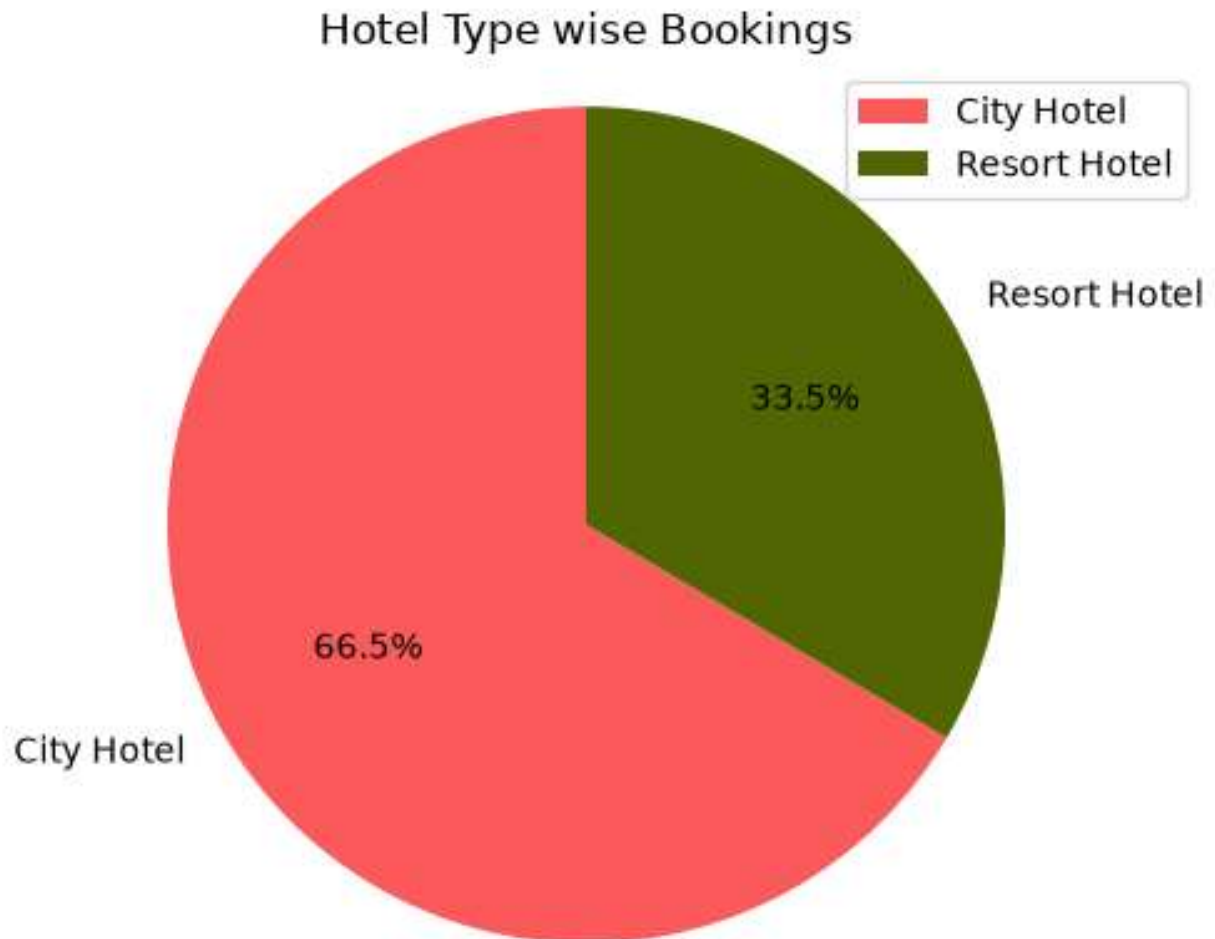


Figure 1. Share of bookings by hotel type.

- **City Hotel takes 66.5% of all bookings (78,370)** versus Resort Hotel's 33.5% (39,483) — City Hotel is booked exactly twice as often.
- This sets the baseline for every later comparison: because City Hotel carries roughly two-thirds of total volume, its behaviour dominates overall dataset trends unless results are split by hotel type.

## 2.2 — What meal plan do guests choose most often?

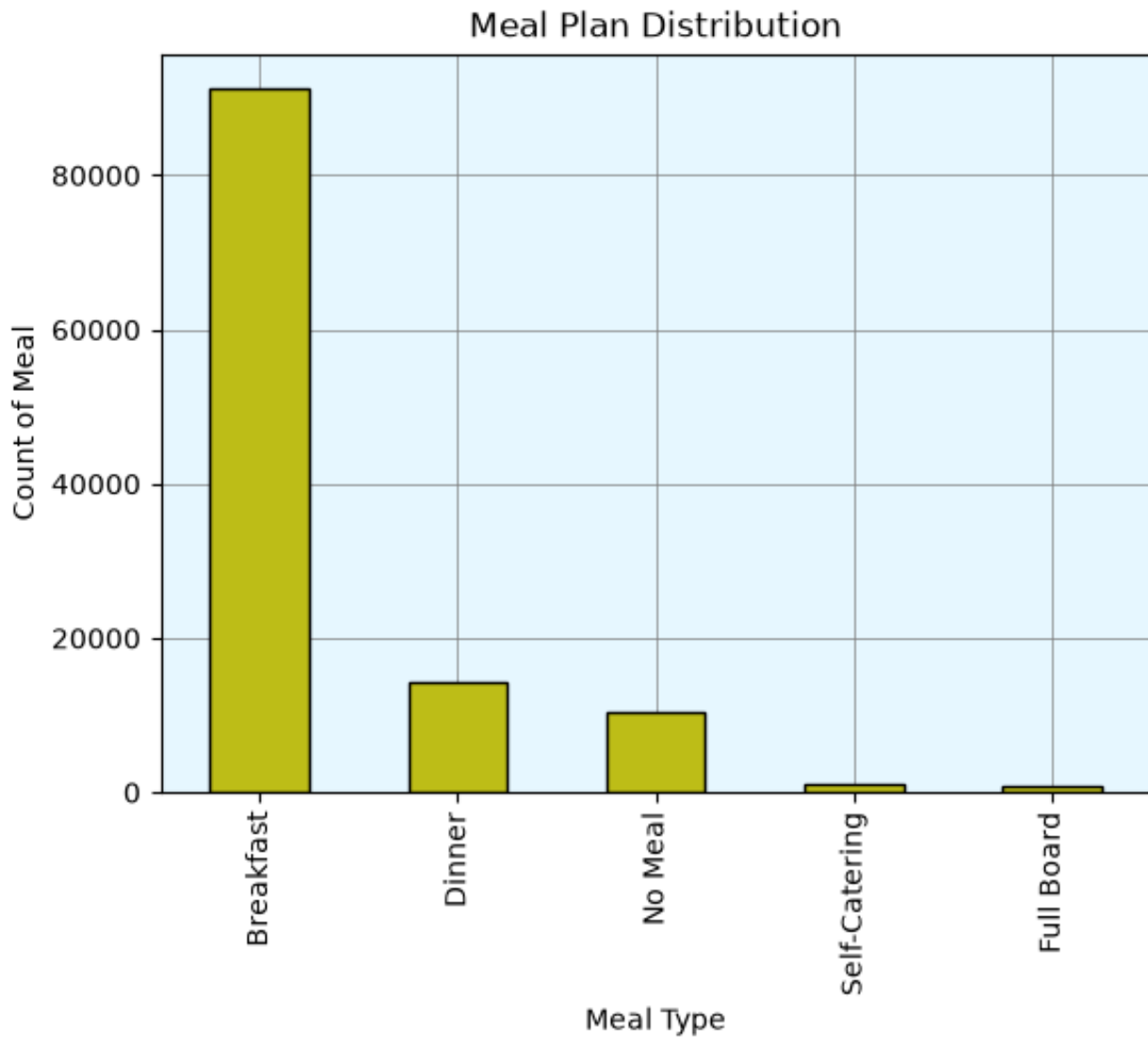


Figure 2. Distribution of meal plan bookings.

- Breakfast (Bed & Breakfast) is by far the most common plan, chosen in ~77.4% of bookings (91,299 bookings).
- Dinner is a distant second (~12.1%), followed by No Meal (~8.9%).
- Self-Catering (recategorized from "Undefined") and Full Board are both under 1% of bookings.
- Takeaway: most guests default to or prefer a bundled bed-and-breakfast package over self-catering or full-board options — a useful signal for how food & beverage packages should be marketed and priced.

### 2.3 — What does the distribution of Average Daily Rate (ADR) look like?

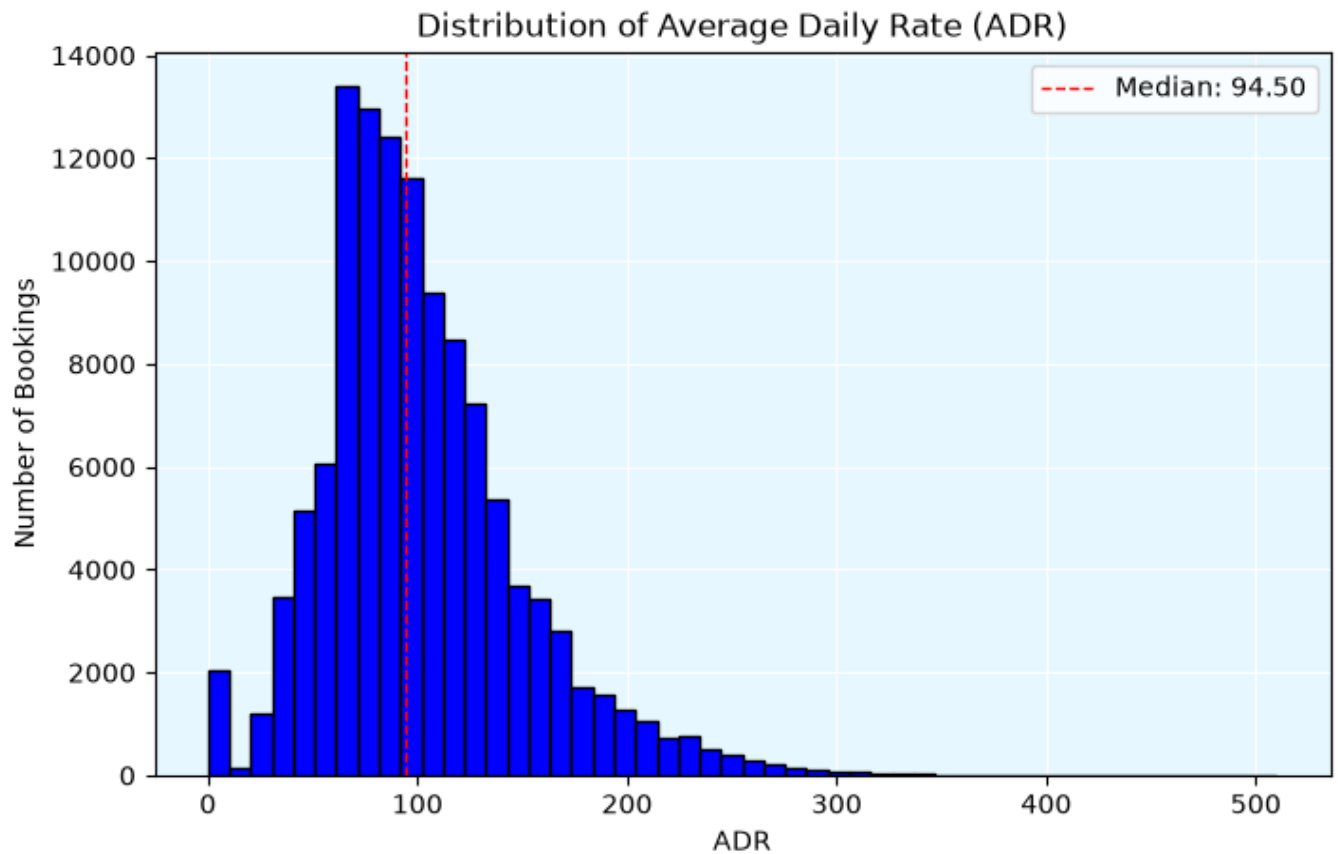


Figure 3. Distribution of Average Daily Rate (ADR), post-cleaning.

- Median ADR is 94.5; the middle 50% of bookings (25th–75th percentile) fall between 69 and 125.8 per night.
- Mean ADR (101.6) sits slightly above the median, confirming a right-skewed distribution with a tail of higher-priced bookings pulling the average up.
- After cleaning, ADR ranges realistically from 0 to 510, with no more negative values or extreme 5,400-type outliers.
- Overall, pricing is fairly consistent for most stays, with only a small share of premium-priced bookings.

### 3. Bivariate Analysis

Pairwise relationships between key drivers and cancellation outcome.

#### 3.1 — Does cancellation rate differ between hotel types?

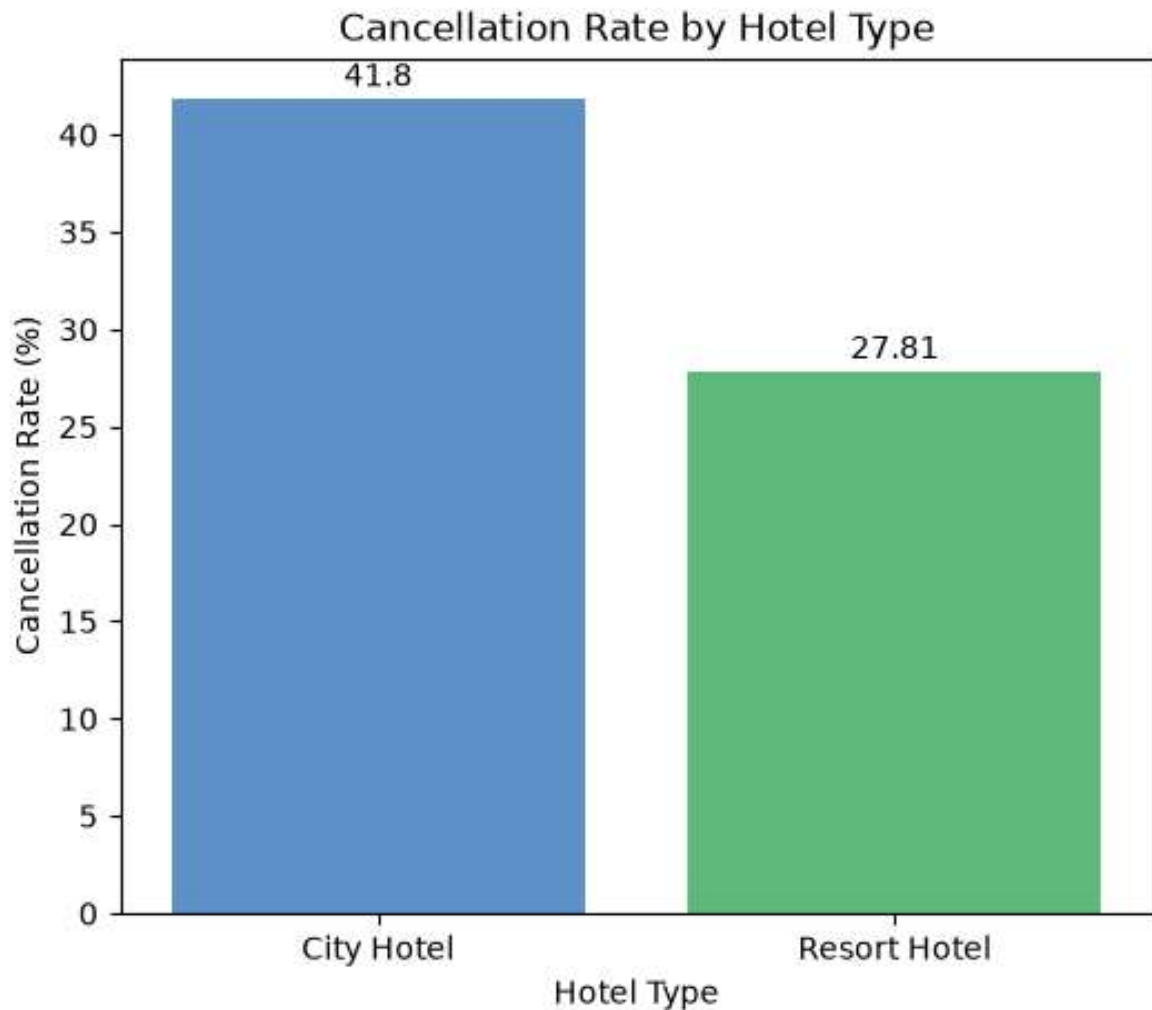


Figure 4. Cancellation rate by hotel type.

- **City Hotel cancels at ~42%**, versus ~28% for Resort Hotel — a gap of roughly 14 percentage points.
- Plausible drivers: more flexible booking/cancellation terms, business-travel plans changing at short notice, and more price-sensitive guests willing to rebook elsewhere.
- Resort Hotel's lower rate likely reflects more vacation-planned, advance-committed bookings.

### 3.2 — Does deposit type affect cancellation rate?

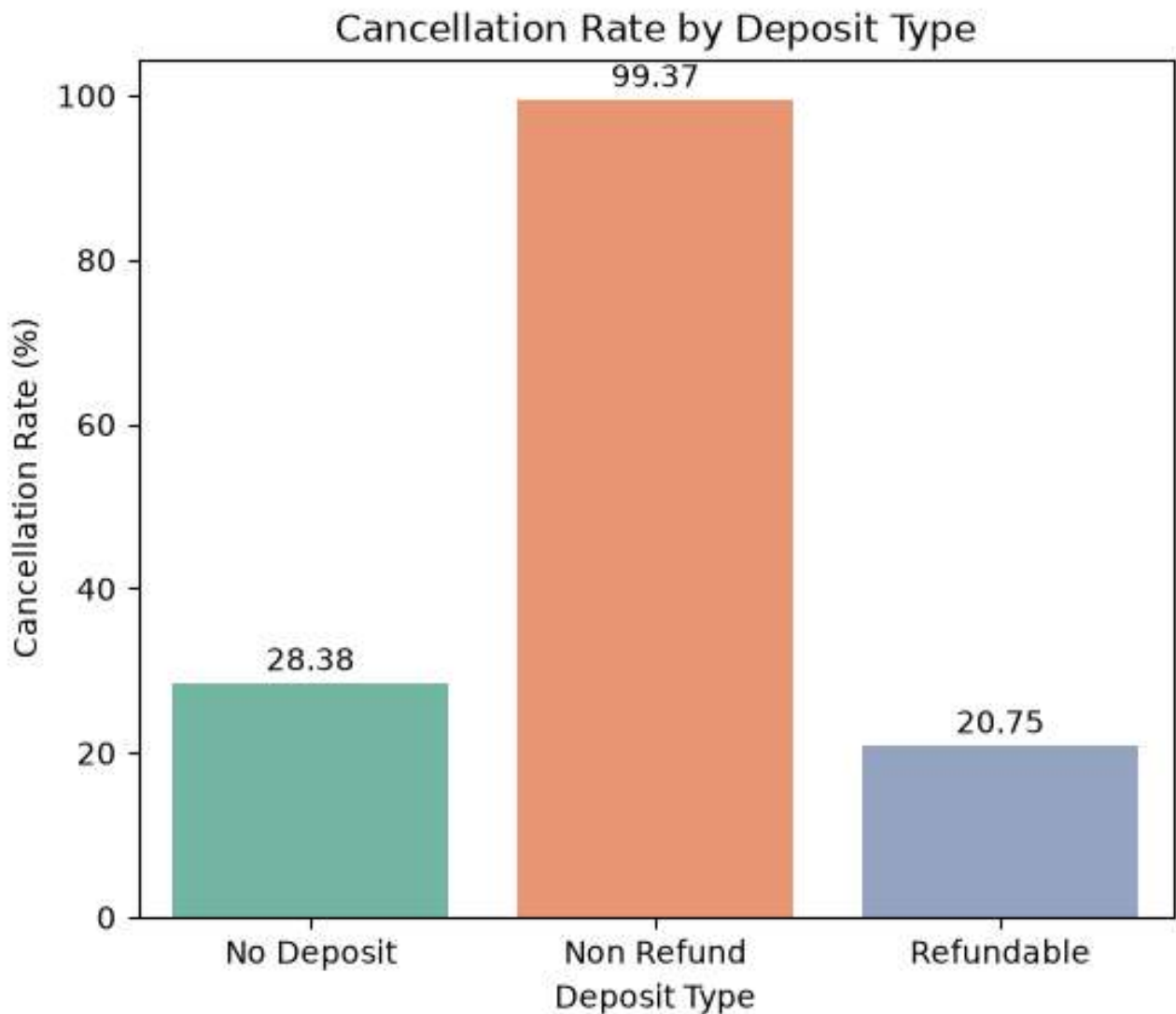


Figure 5. Cancellation rate by deposit type.

- **Non Refund bookings cancel at 99.37%** — almost every single one — which is highly counter-intuitive, since guests forfeit money by cancelling.
- No Deposit bookings cancel at 28.38%; Refundable bookings cancel least, at 20.75% (though this group is small — only 159 bookings — so its rate is less statistically reliable).
- Likely explanation: the Non Refund spike does not represent guests knowingly forfeiting a deposit — it more plausibly reflects no-shows, agent/wholesaler bulk holds, or a data-labeling quirk. This is a channel to investigate and monitor, not a proven cancellation deterrent.

### 3.3 — Are repeat guests less likely to cancel?

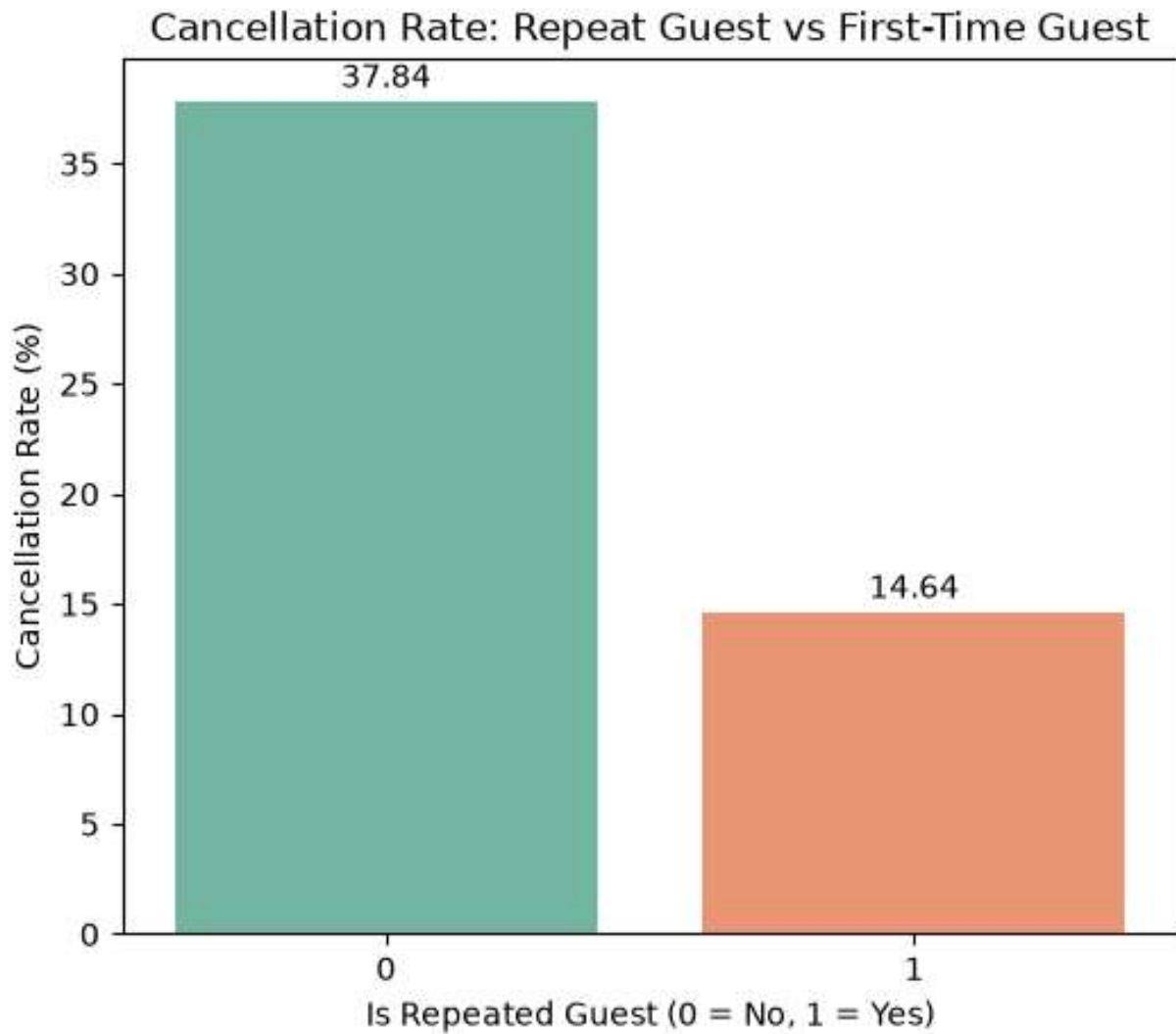


Figure 6. Cancellation rate: repeat vs first-time guests.

- **First-time guests cancel at 37.84%; repeat guests cancel at just 14.64% — less than half the rate.**
- Repeat guests make up only ~3.2% of all bookings (3,715 of 117,853), but are by far the most reliable segment — a strong case for investing in loyalty and retention programs.



## 4. Multivariate Analysis

Combined views across hotel type, time, and pricing to explain the seasonal and behavioural patterns behind the headline numbers.

### 4.1 — How do monthly bookings compare across hotel types?

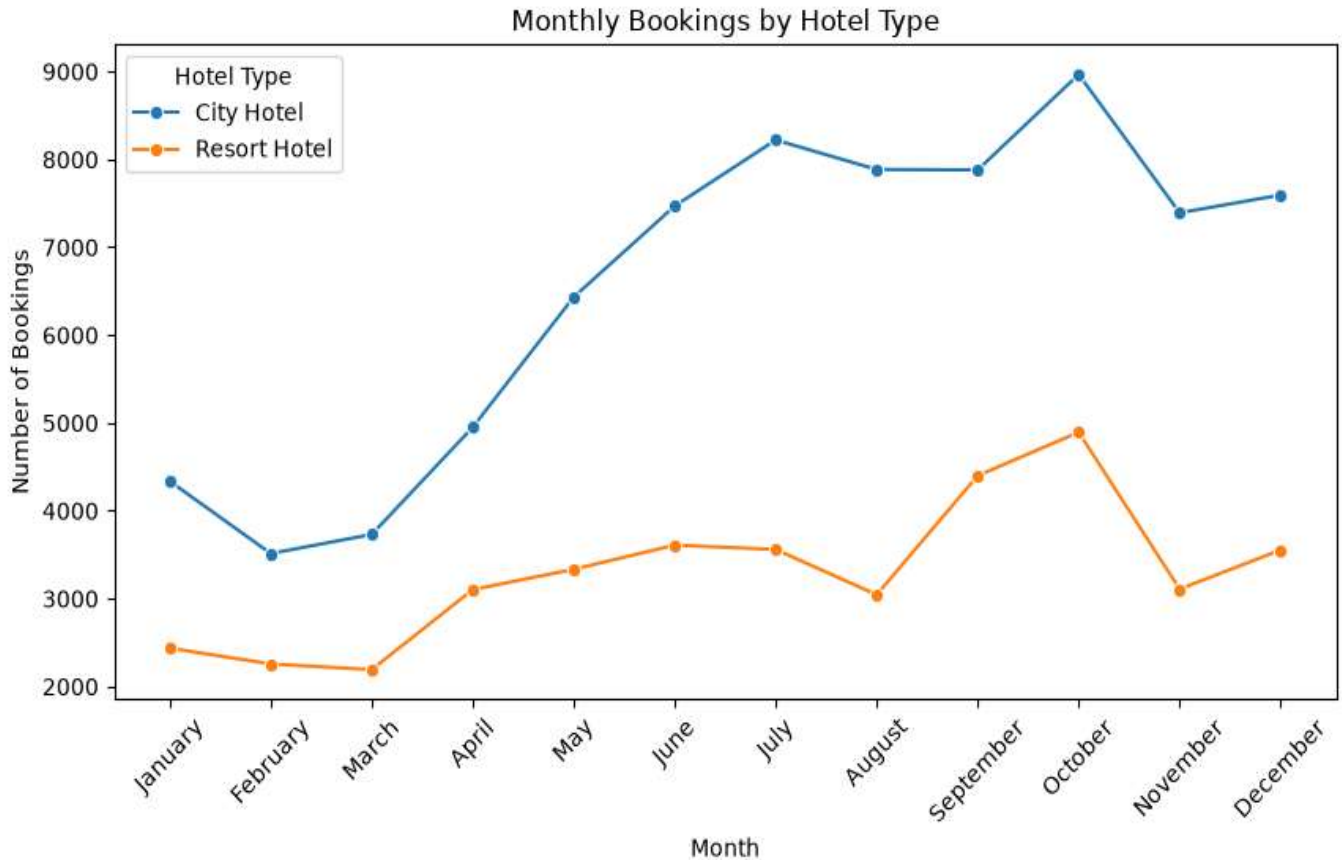


Figure 7. Monthly bookings by hotel type.

- October is the peak month for both hotel types.
- City Hotel's low point is February; Resort Hotel's low point is March.
- City Hotel outsells Resort Hotel in every single month, showing steadier, less seasonal demand, while Resort Hotel bookings are more tied to vacation season, surging through summer into autumn.
- Operational takeaway: use the January–March trough for maintenance and off-season promotions, and staff up ahead of the October peak.

#### 4.2 — Does length of stay affect cancellation rate, by hotel type?

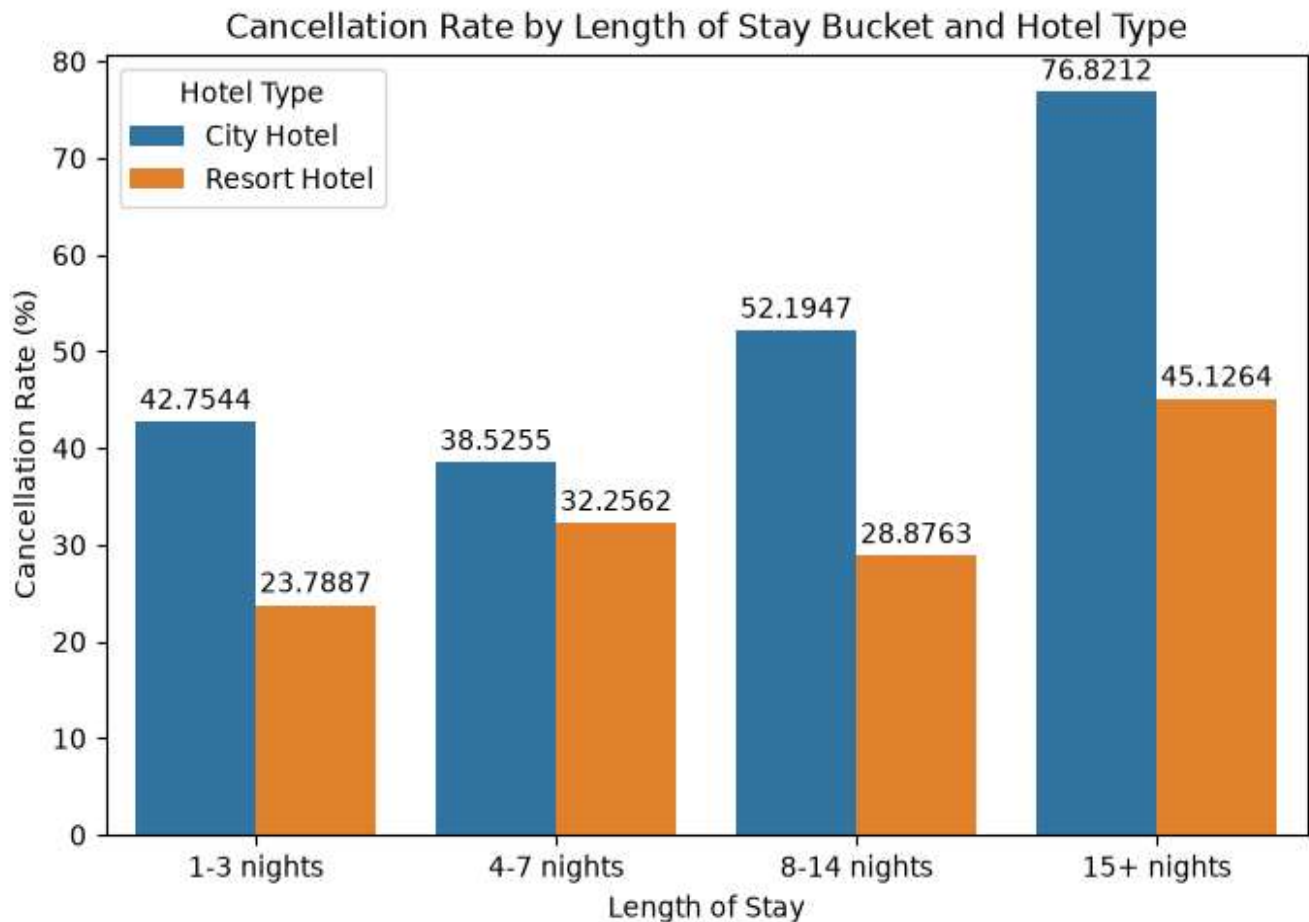


Figure 8. Cancellation rate by length-of-stay bucket and hotel type.

- **Cancellation rate rises with stay length for both hotel types**, but far more sharply for City Hotel: 42.75% (1–3 nights) climbing to 76.82% (15+ nights).
- Resort Hotel rises far more gently: 23.79% (1–3 nights) to 45.13% (15+ nights).
- 15+ night bookings are rare (151 City, 277 Resort out of 117,853 rows), so this segment's rate is less statistically stable than shorter stays — worth flagging when acting on it.
- Longer bookings likely carry more planning uncertainty (schedule or budget changes), and City Hotel's more flexible, business-driven guest mix appears quicker to cancel than Resort Hotel's vacation-committed guests.

### 4.3 — Does lead time affect cancellation rate, by hotel type?

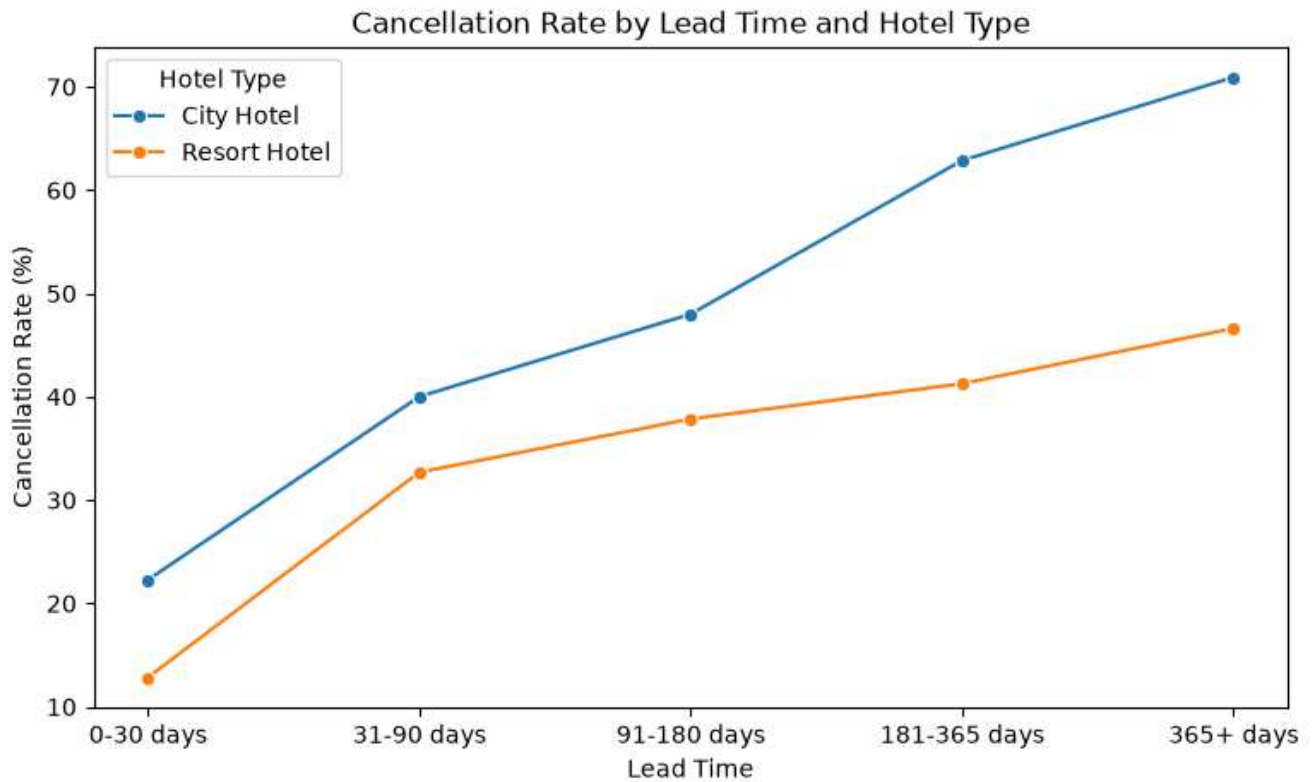


Figure 9. Cancellation rate by lead-time bucket and hotel type.

- **Cancellation climbs steadily with lead time for both hotel types.** City Hotel rises from 22.28% (0–30 days) to 70.92% (365+ days); Resort Hotel from 12.83% to 46.62%.
- The lowest cancellation rate for both hotel types is for bookings made within 30 days of arrival — last-minute guests are far more committed.
- City Hotel is consistently and substantially higher than Resort Hotel at every lead-time bracket, and the gap widens the further out the booking is made.

#### 4.4 — How does ADR change across months, by hotel type?

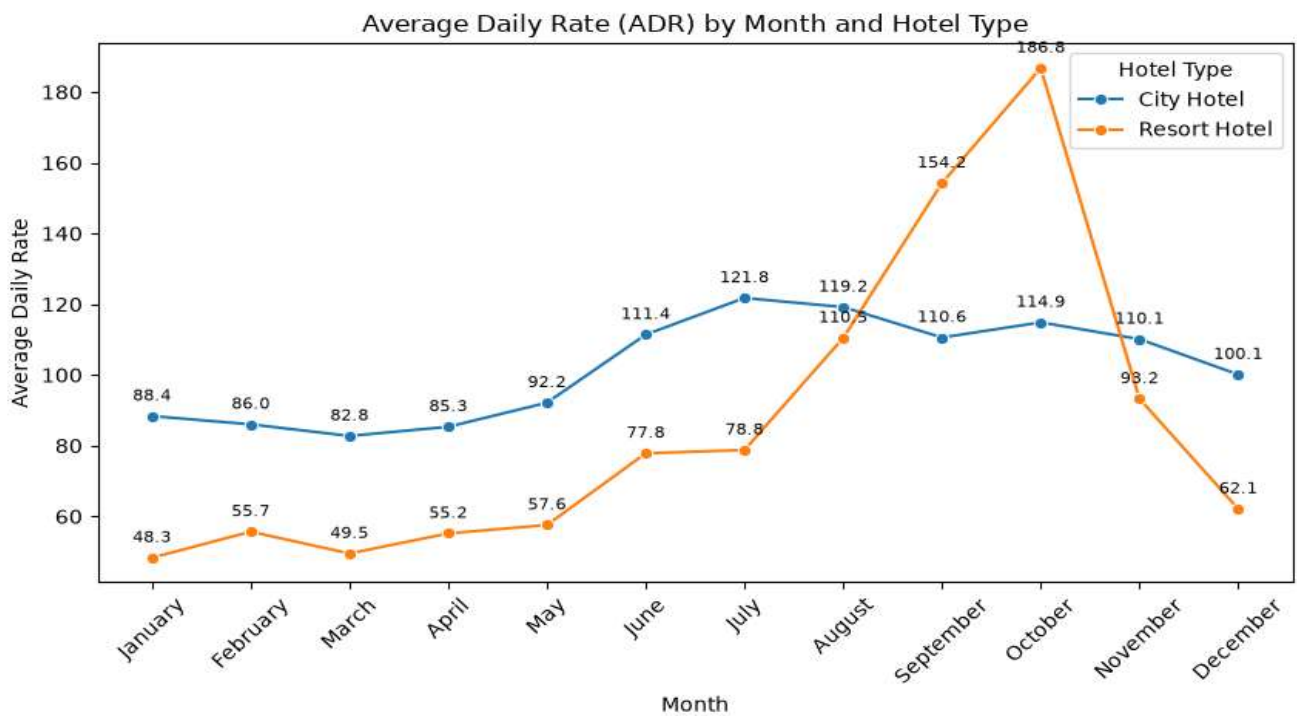


Figure 10. Average Daily Rate by month and hotel type.

- City Hotel's ADR stays relatively stable year-round, ranging from ~83 (March) to ~122 (July) — a fairly flat curve.
- Resort Hotel's ADR swings dramatically, from a low of ~48 (January) to a peak of ~187 (October), overtaking City Hotel entirely in September–October.
- This is a different dynamic from the booking-volume peak seen in Figure 7 (also October for both): Resort Hotel doesn't just attract more bookings in autumn, it also charges significantly more per night, pointing to deliberate seasonal demand pricing rather than volume alone.

#### 4.5 — Which market segment cancels most, by hotel type?

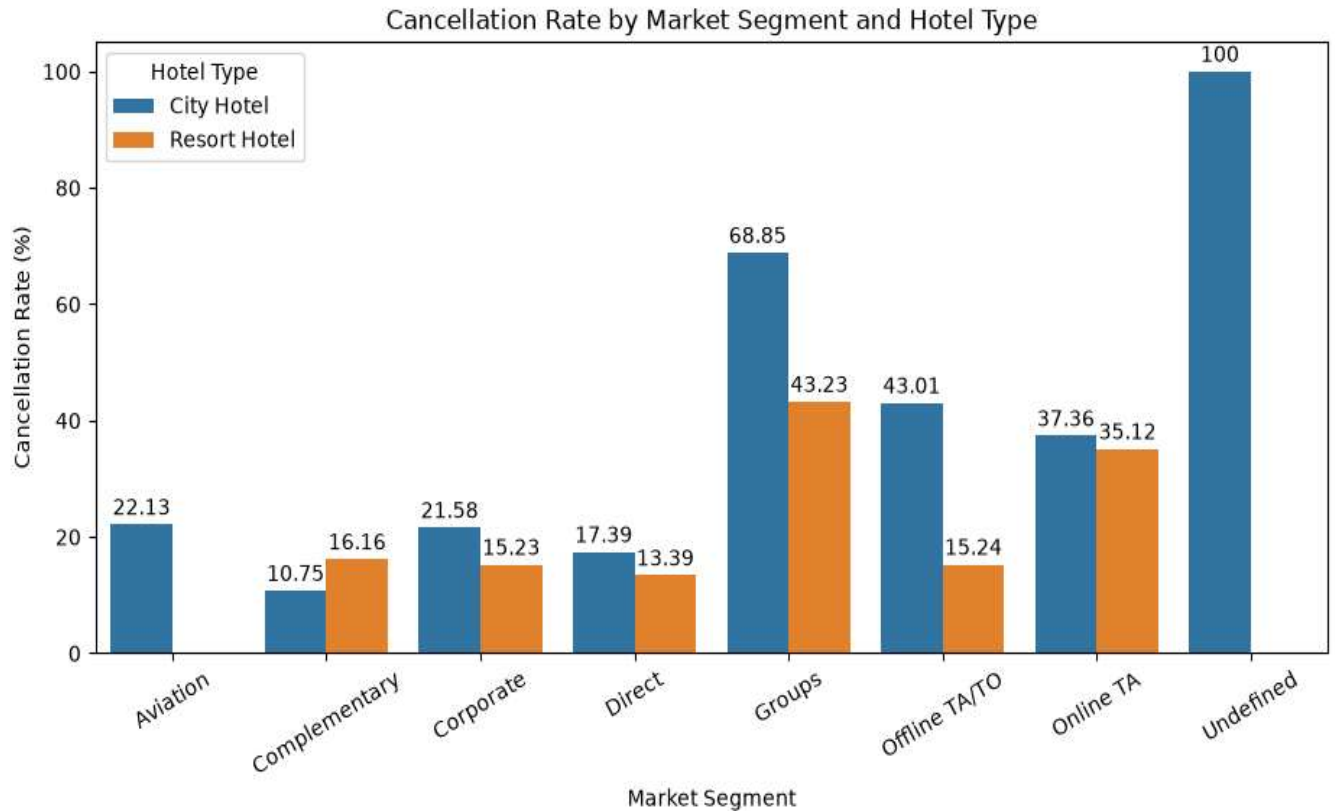


Figure 11. Cancellation rate by market segment and hotel type.

- **Groups is the riskiest segment for City Hotel (68.85% cancellation),** far above Resort Hotel's Groups rate (43.23%).
- Complementary (free/comp) stays cancel the least for City Hotel (10.75%) — consistent with the earlier zero-ADR finding that comp bookings are the most reliable.
- Offline TA/TO shows a striking gap: City Hotel cancels at 43.01% versus Resort Hotel's 15.24% — the same booking channel behaves very differently depending on hotel type.
- City Hotel cancels more than Resort Hotel in nearly every segment, but Groups and Offline TA/TO show by far the widest gaps — these are the segments driving City Hotel's overall higher cancellation rate.

#### 4.6 — Does ADR vary with lead time, by hotel type?

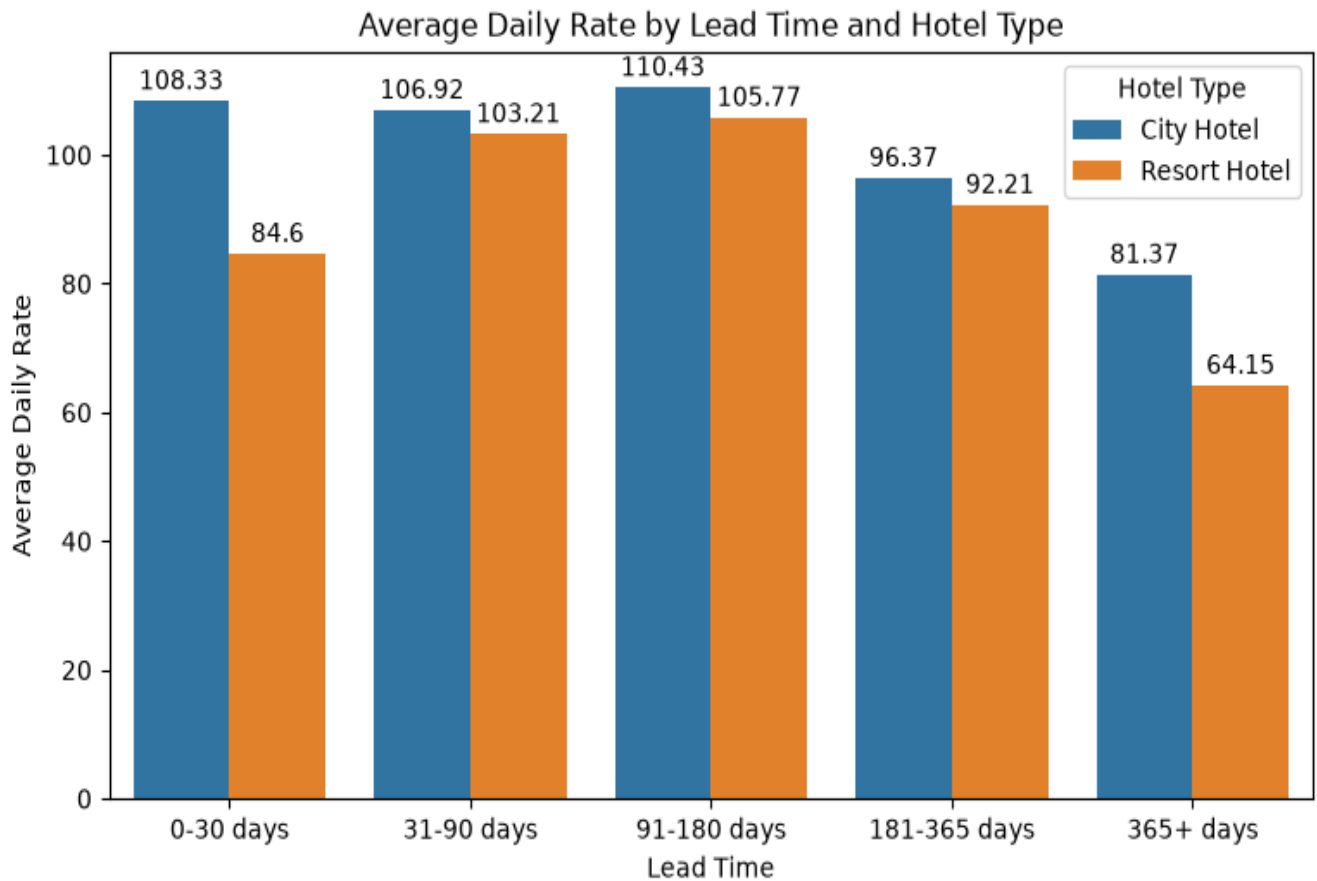


Figure 12. Average Daily Rate by lead-time bucket and hotel type.

- For City Hotel, ADR stays fairly steady across most lead-time brackets (~106–110), dropping only for very early bookings (365+ days: 81.37).
- Resort Hotel shows a clearer rise-then-fall pattern: lowest at 0–30 days (84.60), rising to a peak around 91–180 days (105.77), then dropping sharply for 365+ day bookings (64.15).
- Both hotel types charge their lowest average rates to guests booking 365+ days out, consistent with early-bird discount pricing to lock in demand far ahead of time.
- Resort Hotel's last-minute guests pay noticeably less than City Hotel's — a likely "fill the room" discounting strategy for less predictable, more seasonal demand.

## 5. Conclusion

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This EDA analysed 117,853 hotel bookings (2017–2019) to answer three core business questions: which hotel type is booked more often, whether stay duration affects cancellation, and whether lead time affects cancellation. Across nearly every cut of the data — stay length, lead time, deposit type, and market segment — City Hotel is more popular but consistently cancels more than Resort Hotel, and that gap widens under higher-risk conditions such as longer stays, longer lead times, and Group or Offline TA/TO bookings.

### Hotel-type popularity

City Hotel is booked twice as often as Resort Hotel (66.5% vs 33.5% of bookings) and holds steadier demand across the year. Resort Hotel is more seasonal, with a pronounced vacation-driven peak in September–October, when its ADR also spikes well above City Hotel's.

### Stay duration and cancellation

Cancellation rate rises with length of stay for both hotel types, but City Hotel is far more sensitive — climbing from 42.75% (1–3 nights) to 76.82% (15+ nights), compared with Resort Hotel's gentler rise from 23.79% to 45.13%.

### Lead time and cancellation

Cancellation rate rises steadily the further in advance a booking is made. City Hotel again shows the sharper increase, from 22.28% (0–30 days) to 70.92% (365+ days), compared with Resort Hotel's 12.83% to 46.62%.

## 6. Recommendations

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- **Strengthen cancellation policy for long-lead-time bookings.** Cancellation risk rises steadily and substantially the further out a booking is made, especially past 90–180 days. Consider partial non-refundable deposits, confirmation reminders closer to arrival, or graduated cancellation fees for bookings made many months in advance — particularly at City Hotel.
- **Target City Hotel's highest-risk segments directly.** Groups (68.85% cancellation) and Offline TA/TO (43.01%) bookings at City Hotel cancel far more than the same segments at Resort Hotel. These channels warrant tighter deposit or confirmation requirements.
- **Apply minimum-stay or scaled-deposit rules for longer bookings.** Stays of 8+ nights — especially at City Hotel, where cancellation reaches 52–77% — carry disproportionate cancellation risk. A deposit that scales with stay length could offset lost revenue.
- **Investigate the Non Refund deposit anomaly.** A 99.37% cancellation rate on Non Refund bookings is very unlikely to reflect guests knowingly forfeiting money — it more plausibly reflects no-shows or agent/wholesaler bulk holds. This channel needs closer monitoring rather than being treated as a proven cancellation deterrent.
- **Lean into Resort Hotel's autumn pricing power, and test the same at City Hotel.** Resort Hotel's ADR climbs sharply in September–October alongside its highest booking volume of the year, showing real pricing power during peak demand. A similar dynamic-pricing approach could be trialled in City Hotel's own peak months to capture more revenue.
- **Invest in repeat-guest retention.** Repeat guests cancel at less than half the rate of first-timers (14.64% vs 37.84%) but are only ~3.2% of bookings — loyalty incentives could grow this highly reliable segment.

### Highest-Impact Recommendation

Tightening cancellation policy for long-lead-time bookings is likely the single highest-impact change. Lead time affects the largest share of bookings across both hotel types, and shows the most consistent, steep relationship with cancellation of any factor examined in this analysis — meaning even a modest reduction in far-ahead cancellations would translate into a meaningful reduction in lost revenue.